Time series: spectrums and filtering

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The homework is provided on the last slide

Karhunen-Loeve and Mercer expansions

Theorem

Let x_t be a stochastic process defined on the interval $[a, b]$ such that $\mathbf{E}x_t = 0$ and the covariance function $R(t, s) = \mathbf{E}x_t x_s$ is continuous. Then there is the **Mercer expansion** of the covariance function

$$R(t, s) = \sum_{j=1}^{\infty} \lambda_j \phi_j(t) \phi_j(s),$$

where $\phi_1(t), \phi_2(t), \ldots$ form an orthonormal basis of $L_2([a, b])$. Also, there is the **Karhunen-Loeve expansion** of the stochastic process

$$x_t = \sum_{j=1}^{\infty} \epsilon_j \sqrt{\lambda_j} \phi_j(t)$$

where $\{\epsilon_j\}_{j=1,2,\ldots}$ is white noise with unit variance.

Bochner-Khinchin theorem, continuous case

Theorem

Let x_t be a **weakly stationary continuous** process, $\mathbf{E}x_t = 0$, $R(t-s) = \mathbf{E}x_t x_s$. Then there is the spectral representation

$$R(t) = \int_{-\infty}^{\infty} e^{it\lambda} dF(\lambda)$$

where $F(\lambda)$ is called a **spectral distribution**.
If $F(\lambda)$ is absolutely continuous, then we have

$$R(t) = \int_{-\infty}^{\infty} e^{it\lambda} f(\lambda) d\lambda$$

where $f(\lambda)$ is called a **spectral density** (a **spectrum**).

The process has the spectral representation $x_t = \int_{-\infty}^{\infty} e^{it\lambda} d\mu(\lambda)$, where $i = \sqrt{-1}$ and $e^{it\lambda} = \cos(\lambda t) + i \sin(\lambda t)$.

Bochner-Khinchin theorem, discrete case

Theorem

Let x_k be a **weakly stationary discrete** process, $\mathbf{E}x_k = 0$, $R(k-m) = \mathbf{E}x_k x_m$. Then there is the spectral representation

$$R(k) = \int_{-\pi}^{\pi} e^{ik\lambda} dF(\lambda)$$

where $F(\lambda)$ is called a **spectral distribution**.
If $F(\lambda)$ is absolutely continuous, then we have

$$R(k) = \int_{-\pi}^{\pi} e^{ik\lambda} f(\lambda) d\lambda$$

where $f(\lambda)$ is called a **spectral density** (a **spectrum**).

In the discrete case, integrals have different limits of integration. The process has the spectral representation $x_k = \int_{-\pi}^{\pi} e^{ik\lambda} d\mu(\lambda)$.

Interpretation by example

The spectral representation can be explained by example: many devices (radio, TV, mobile phone, WiFi) work with signals in the form of waves based on mathematical principles.





Loudspeaker and Microphone Design

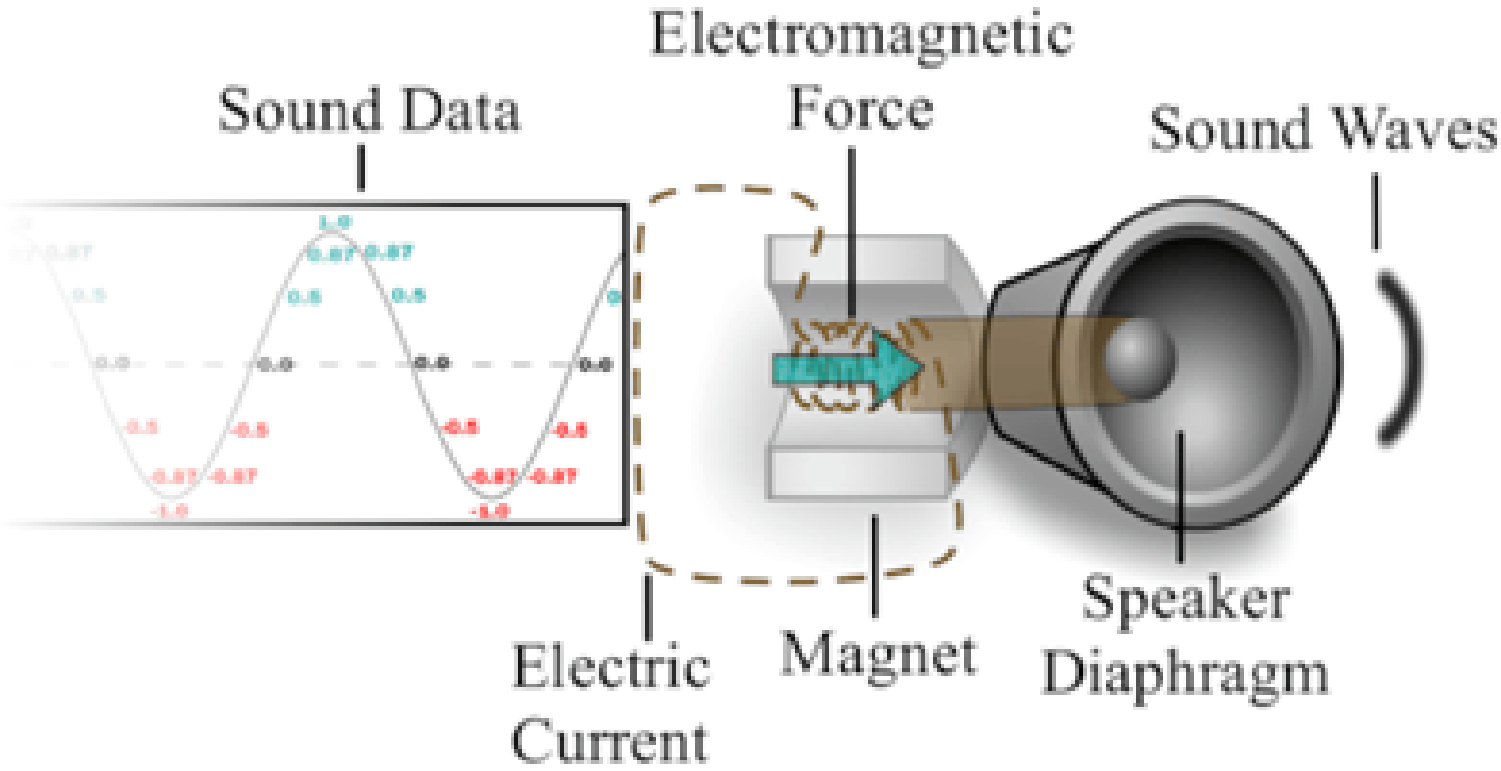
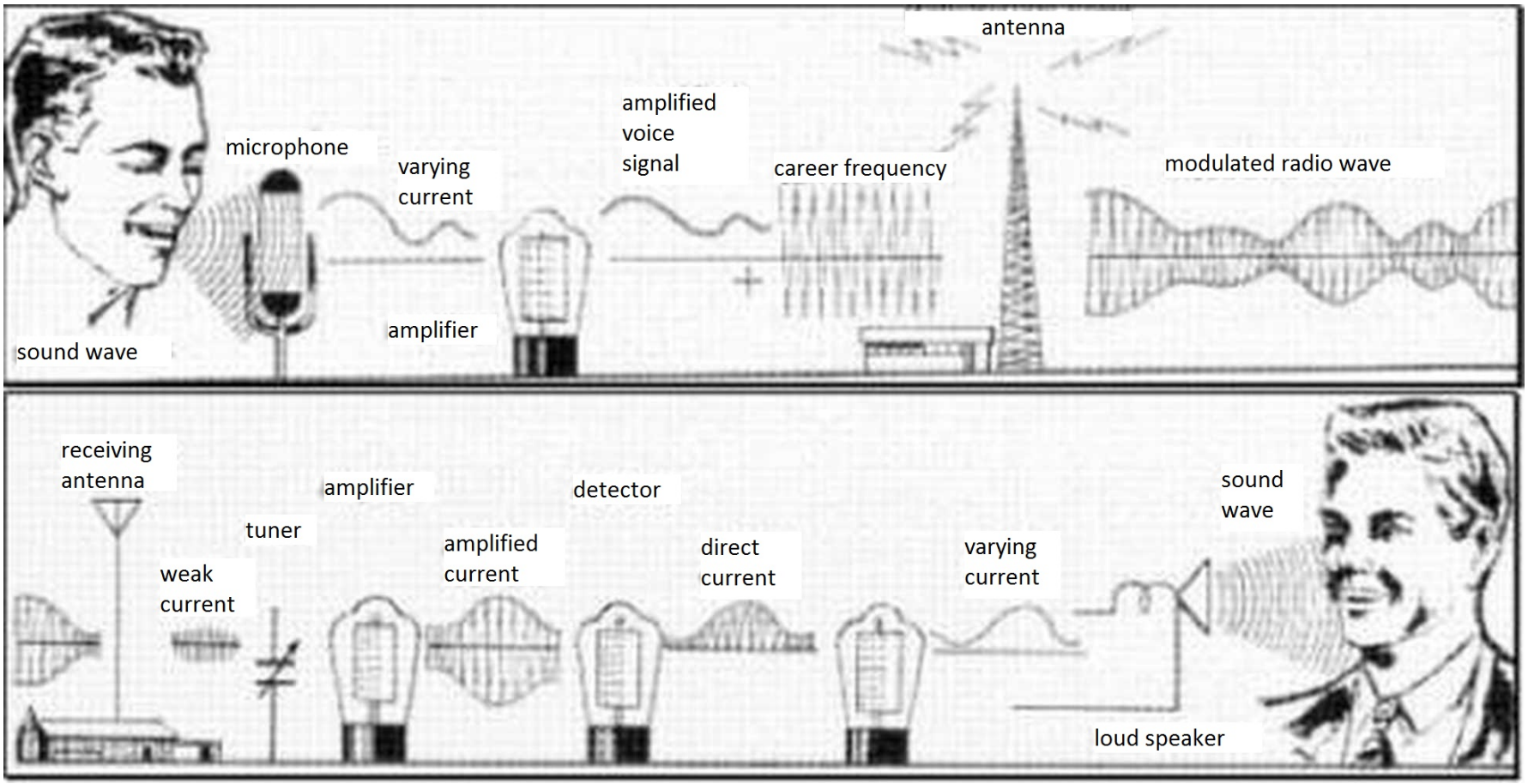
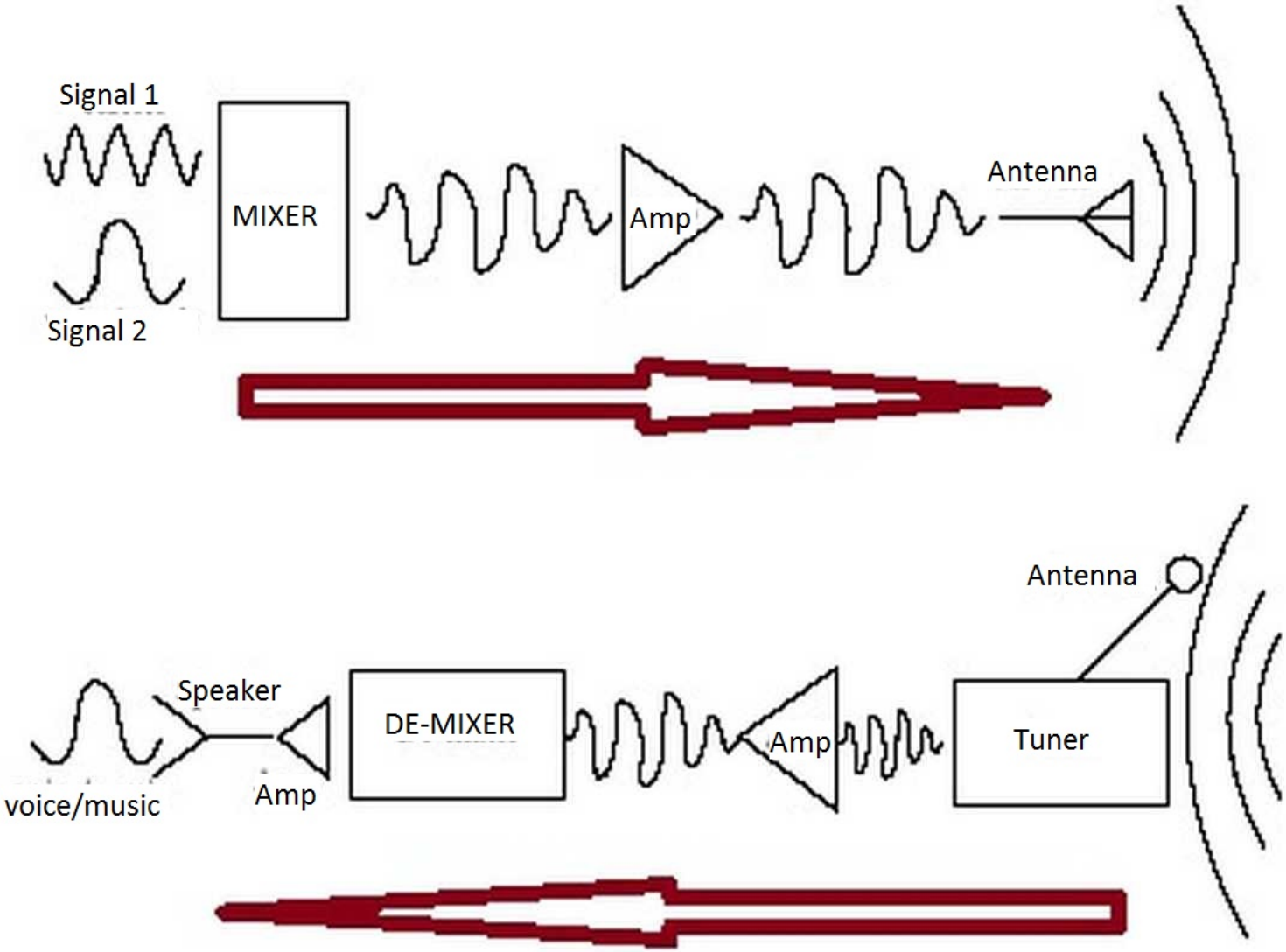


Diagram showing how sound data turns into sound waves: A computer representation of sound data is turned into an electrical current, which creates an electromagnetic force, which moves a speaker diaphragm, which creates sound waves.
`https://en.wikipedia.org/wiki/Loudspeaker`

Transmission of a signal



Mixing and de-mixing of a signal



Mixing: sum of sinusoids

A sum of sinusoids

$$\sum_{j=1}^{m} c_i \sin(a_i t)$$

results in a wave with a complex form.

® *n=1000; a1=1.3; a2=2.9; tm=seq(-50, 50, length.out=n);*
*plot(tm,sin(a1*tm)+sin(a2*tm), type="l",*
main="sin1+sin2(black), sin1(blue), sin2(red)", col="black");
*lines(tm, sin(a1*tm), col="blue");*
*lines(tm, sin(a2*tm), col="red")*

Temporal and spectral representation

Any wave has two representations:

- **Temporal representation**: a time series x_k obtained by sampling the wave at equal time intervals.
- **Spectral representation**: the spectrum computed using the Bochner-Khinchin theorem for x_k.

The spectrum of a sinusoid $x_k = \sin(ak)$, $k \in \mathbb{Z}$, is the Dirac delta function at the point a, i.e. $f(\lambda) = \delta_a(\lambda)$, $\lambda \in [0, \pi]$.

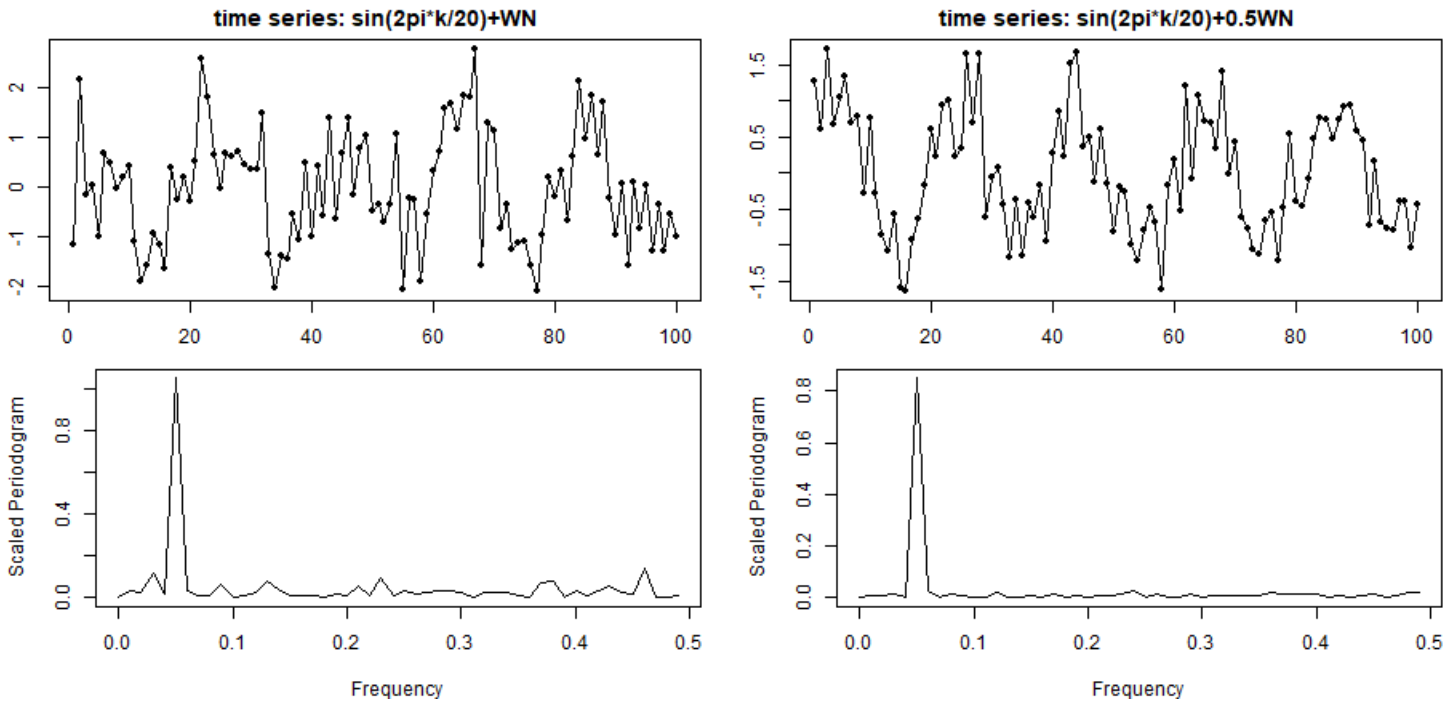
The **spectrum of the sum** of waves is the **sum of their individual spectrums**.

λ is called an **angular frequency**.
$\nu = \frac{\lambda}{2\pi}$ is called an **ordinary frequency**.

Numerical Exercise: sketch

Task: Sketch the temporal and spectral representations of $x_k = \sin(2\pi k/20) + \epsilon_k$ and $y_k = \sin(2\pi k/20) + 0.5\varepsilon_k$, where ϵ_k and ε_k are Gaussian white noise with zero mean and variance 1. Explain any differences in your sketches.

Solution:



The series x_k has a higher noise level. The periodogram for x_k has a peak at the same frequency as the periodogram for y_k. However, the periodogram for x_k has larger fluctuations at other frequencies.

Filtering / de-mixing

'Filtered series'=Filter('Base series')

Any filter has two representations:

- **Temporal representation**: a sequence of coefficients $\ldots, q_{-2}, q_{-1}, q_0, q_1, q_2, \ldots$.
- **Spectral representation**: the spectrum computed using the Bochner-Khinchin theorem for q_k.

The **spectrum of the filtered series** is the **product of the spectrums** of the filter and the base series.

A common choice for the filter spectrum is the indicator function, $f_q(\lambda) = 1_{[b_1,b_2]}(\lambda)$.

Numerical Exercise: filtering

Task: Consider the time series
$x_k = \sin(2\pi k/25) + 2\sin(2\pi k/15) + 3\sin(2\pi k/9),\ k \in \mathbb{Z},$
and the filter with spectrum $f(\lambda) = 1_{[0.1\pi, 0.2\pi]}(\lambda),\ \lambda \in (0, \pi)$.
Find the filtered series and its spectrum.

Solution: The spectrum of x_k can be written as
$f_x(\lambda) = \delta_{2\pi/25}(\lambda) + 2^2\delta_{2\pi/15}(\lambda) + 3^2\delta_{2\pi/9}(\lambda)$.
The spectrum of the filtered series is the product of the spectrum of x_k and the spectrum of the filter. Since $2\pi/25 \notin [0.1\pi, 0.2\pi]$, $2\pi/15 \in [0.1\pi, 0.2\pi]$, $2\pi/9 \notin [0.1\pi, 0.2\pi]$, we have that the spectrum of the filtered series is $f_y(\lambda) = f_x(\lambda)f(\lambda) = 2^2\delta_{2\pi/15}(\lambda)$ and the filtered series is given by $y_k = 2\sin(2\pi k/15)$.

Convolution

The operation

'Filtered series'=Filter('Base series')

takes the form of a convolution, that is,

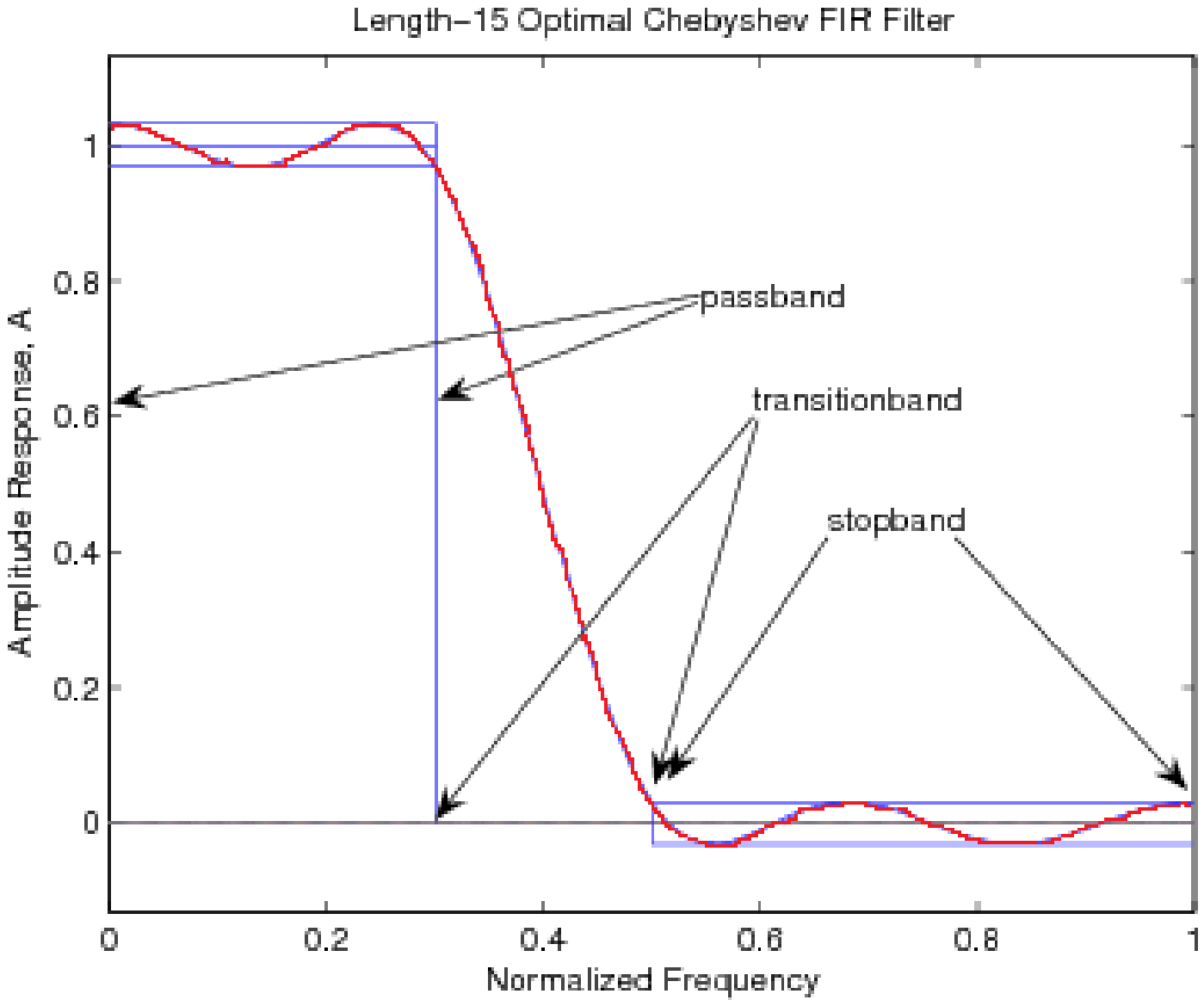
$$y_k = \sum_{j=-\infty}^{\infty} q_j x_{k-j}.$$

where y_k is the filtered series and x_t is the base series.

In practice, a filter with an infinite number of coefficients cannot be implemented.

By applying truncation, we obtain a filter with a small number of coefficients, though its spectrum becomes imperfect.

Ideal and approximating filters



The ideal filter has the spectrum $f_q(\lambda) = 1_{[0,0.3]}(\lambda)$.

Homework

Apply AI tools to deepen your understanding of spectrums and filtering.

- **Karhunen-Loeve Expansion:** Find the specific eigenvalues λ_j and eigenfunctions $\phi_j(t)$ for a standard Brownian Motion on the interval $[0, 1]$.
- **Simulation in R:** Generate a signal x_k consisting of three sinusoids with different frequencies and add Gaussian noise. Plot the temporal series and its estimated spectral density (periodogram).
- **Filtering Task:** Write R code to implement a simple moving average filter as a convolution. Observe how this "low-pass" filter affects the spectrum of a noisy signal.
- **Filter Truncation:** Investigate what happens to the frequency response of an "ideal" filter when you truncate its infinite series of coefficients to a small number.
- Solve exam-like exercises for L5.